Liou Tang

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lioutang.github.io

RESEARCH INTEREST

Privacy-Preserving Machine Learning ○ Machine Unlearning ○ Quantum Machine Learning

EDUCATION

University of Pittsburgh *Sep. 2024 -*
Ph.D. in Information Science; Advised by Dr. James Joshi

University of Pittsburgh *Sep. 2022 - Apr. 2024*
Master of Science in Information Science

Shanghai Jiao Tong University *Sep. 2018 - Jun. 2022*
Bachelor of Engineering in Information Security

GRANTS AND PROJECTS

*: Under supervision of Dr. James Joshi as principle investigator.

EAGER: Trustworthy and Verifiable Machine Unlearning in an Adversarial Environment *LERSAIS Lab*
Role: Participant *Oct. 2026 - Sep. 2028*

- Funded by NSF Secure and Trustworthy Cyberspace (SaTC) program under award no. 2624954

Quantum Secure and Privacy-Preserving Machine Learning in Hybrid Environments *LERSAIS Lab*
Role: Participant *Jan. 2026 -*

- Funded by Cisco Research.

SecPriMU: Towards A Holistic Secure And Privacy-Preserving Machine Unlearning Framework *LERSAIS Lab*
Role: Participant *Jan. 2025 - Aug. 2025*

- Funded by Cisco Research.

PUBLICATIONS

[1] L. Tang, P. Krishnamurthy, and M. Abdelhakim, "Is machine learning the best option for network routing?" in *ICC*. IEEE, 2024, pp. 5425–5430.

[2] L. Tang and J. Joshi, "Towards privacy-preserving and secure machine unlearning: Taxonomy, challenges and research directions," in *TPS-ISA*. IEEE, 2024, pp. 280–291.

[3] L. Tang, J. Joshi, and A. Kundu, "Apollo: A posteriori label-only membership inference attack towards machine unlearning," *CoRR*, vol. abs/2506.09923, 2025.

[4] P. He, L. Tang, M. A. Rahimian, and J. Joshi, "Conformal-DP: Differential privacy on riemannian manifolds via conformal transformation," *CoRR*, vol. abs/2504.20941, 2025.

[5] L. Tang, J. Joshi, and A. Kundu, "Behavioral audit of machine unlearning has a privacy cost," *CoRR*, vol. abs/2606.14518, 2026.

ACADEMIA SERVICES

Conference Service

- Session Chair for 2025 7th IEEE International Conference on Cognitive Machine Intelligence (CogMI)
- Web Co-Chair for IEEE CIC/TPS/CogMI Co-Located Conferences, 2025-2026

Reviewer for Conferences and Journals

- IEEE Transactions on Dependable and Secure Computing (TDSC)
- IEEE Transactions on Big Data (TBD)
- 2026 8th IEEE International Conference on Cognitive Machine Intelligence (CogMI)
- 2026 28th International Conference on Information and Communications Security (ICICS)
- IEEE 27th International Conference on Information Reuse and Integration for Data Science (IRI)
- 2026 16th ACM Conference on Data and Application Security and Privacy (CODASPY)
- 2025 7th IEEE International Conference on Trust, Privacy and Security in Intelligent Systems, and Applications (TPS-ISA)
- 2024 IEEE 10th International Conference on Network Softwarization (NetSoft)

TEACHING EXPERIENCE

Teaching Assistant

- CS 0449, *Introduction to System Software*, Fall 2025, University of Pittsburgh
- IFSCI 0510, *Data Analysis*, Fall 2024, University of Pittsburgh

Course Design

- ECE 1155, *Information Security*, University of Pittsburgh

INTERNSHIP

Software Engineer Internship at Unity Technologies (Shanghai) *Unity Technologies (Shanghai)*
Role: Software Engineer Intern *Jan. 2022 - Aug. 2022*

- Participated in development of a distributed computing-based cloud gaming system with Entity Component System (ECS).
- Under supervision of Liming Zhang.